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# Agriconnect: Bridging Farms And Bazaars Through Mobile Technology

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***Abstract*— Agricultural supply chains are often inefficient due to the involvement of many middlemen, which reduces farmer revenue and increases consumer expenses. This study proposes a digital network that directly connects farmers and local market sellers, enabling real-time pricing discovery and simplifying supply chain procedures. The solution, which was developed with Spring Boot and SQL for the backend and React Native for the frontend, enables data-driven decision-making and transparent transactions to assist farmers obtain higher pricing while guaranteeing that consumers may purchase reasonably priced, fresh goods. Through automated logistics management, the strategy decreases post-harvest losses and promotes access to agricultural markets. The suggested strategy revolutionizes traditional agricultural trading by utilizing contemporary mobile technologies, giving farmers fair pricing and increasing market efficiency overall.**

***Keywords*— *React Native, Spring Boot, SQL, farmer-market platforms, supply chain optimization, digital agriculture, and real-time pricing discovery*.**

**Introduction**

Inefficiencies caused by the involvement of numerous intermediaries sometimes impede the agricultural supply chain, leading to price discrepancies, lower farmer profit margins, and higher consumer costs [1]. The absence of transparency in traditional trading systems frequently hinders farmers' ability to make informed decisions about pricing and market accessibility [2]. These challenges underline the necessity of a technology-driven solution that raises equality, efficiency, and market visibility.

This research suggests a digital network that links farmers with local market sellers directly in order to cut out needless middlemen and advance a more equitable agricultural trade environment [3]. In order to give farmers access to dynamic market trends and help them make strategic selling decisions, the platform integrates real-time price discovery tools. The suggested solution guarantees scalability, efficiency, and accessibility through an easy-to-use user interface by leveraging React Native for the frontend and Spring Boot with SQL for the backend [4]. This paper highlights the critical role that digital transformation plays in agriculture by highlighting the ways in which technology-driven interventions may empower farmers, improve transparency, and expedite market operations. The ultimate objective is to modernize traditional agricultural trade by providing a farmer-centric, cost-effective, and environmentally friendly strategy for bolstering the agricultural industry.

## I. PROPOSED SYSTEM

To reduce inefficiencies in the agricultural supply chain, a digital network that directly connects farmers with local markets is proposed. In order to improve market accessibility, expedite trade procedures, and guarantee fair pricing, this system incorporates contemporary technical developments. Farmers may access real-time market prices, demand patterns, and trade opportunities using an intuitive smartphone application, enabling them to make well-informed selling decisions [2].

For safe and scalable data administration, the platform makes use of a strong backend constructed using Spring Boot and SQL, and the frontend is made with React Native to provide cross-platform accessibility [4]. In order to give farmers dynamic market information and lessen pricing disparities brought on by middlemen, the platform incorporates real-time price discovery capabilities. Furthermore, by providing price forecasts and demand estimates based on past data and current market patterns, automated analytics and predictive modeling improve decision-making [7]. In conclusion, the suggested automated weed detection and removal system would transform weed management in agriculture by utilizing deep learning and sophisticated robotics. This technique is a major step toward efficient and sustainable farming methods in contemporary agriculture by using artificial intelligence to precisely detect and eradicate weeds while maintaining crop integrity [1].

The suggested digital platform transforms traditional agricultural trading while utilizing technology to advance efficiency, ensure openness, and advance justice. It fosters a more lucrative and sustainable agriculture economy by combining real-time data analytics with an easy-to-use marketplace. In the long run, this increases farmers' financial circumstances and makes the food supply system more resilient [1].

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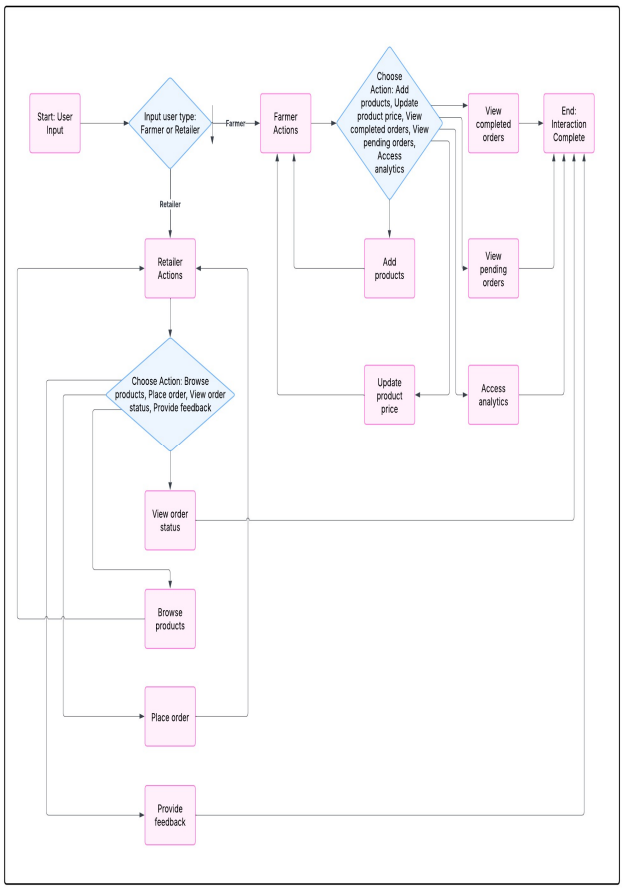



Fig 1. Flow Diagram for Mobile Application

## II. WORKING PROCESS

A local market and farmer connect app guarantees fair prices and easy access to fresh produce by putting farmers in direct contact with wholesalers and retailers [1]. By removing middlemen and lowering transaction costs, such a platform enables farmers to exhibit their goods while retailers can quickly browse and buy straight from the source [2]. Integrated features that help match supply with genuine market demands, such as demand forecasting, safe payment methods, and real-time pricing changes, further improve market efficiency [3]. Because of improved market access and prompt sales, post-harvest losses are greatly decreased, local economies are bolstered, and farmers profit from higher profits [4].

### *A. Planning Phase*

During the planning stage, requirements are gathered and the project's scope is established. Requirements are divided into functional and non-functional categories based on direct interactions with local farmers and market traders (customers).

This project's scope involves creating a system that guarantees a structured and role-specific user experience while facilitating direct transactions between farmers and retailers. This system will include integrated payment mechanisms, a separate login process for farmers and merchants, inventory management for farmers, and product browsing for retailers. Spring Boot for backend services, React Native for frontend development, SQL for database administration, and API integration for smooth component communication are among the technologies chosen for implementation.

### *B. Analysis and Design*

The target users were given structured questionnaires to complete in order to collect data for this phase. Key system functionalities and architectural design were defined by analyzing the gathered data. During the system design process, a number of diagrams are produced, including database schemas, sequence diagrams, activity diagrams, use case diagrams, and architectural diagrams [3]. Because of its interactive prototype characteristics, the Figma tool was utilized for frontend user interface design. A logical diagram is used in the architectural design to illustrate the general system structure and its constituent parts.

### *C. Comparison of Existing Systems and the Proposed Application*

| Evaluation Criteria | Farmer Direct App | Farmer's Market App | Farmers Market App | Agriconnect (Proposed App) |
|---|---|---|---|---|
| Login system | A normal login system that does not differentiate between farmer and consumer | A normal login system that does not differentiate between farmer and consumer, comes with a "Continue as Guest" feature | A normal login system that does not differentiate between farmer and consumer | A separate login system for each user type to differentiate between farmer and customer |
| User type registration | Registers with mobile number but does not differentiate between user types | Normal registration system | Normal registration system | Separate registration to differentiate between local farmers and customers |
| Search tool feature | Provides search feature | Provides search feature | Provides search feature | Provides search feature |
| Farmer's product detail | Provides farmer's product details | Does not provide farmer's product details | Provides farmer's product details | Provides farmer's product details |
| Purchasing method | Purchase system enabled at available farmer's store | Does not support purchasing before checkout | Consumers need to go to the farmer's location to acquire the product | Accepts Google Pay (GPay) and Cash on Delivery (COD) |
| Ability to select product quantity | Able to select product quantity easily | Unable to select product quantity before adding to cart | Unable to select product quantity before adding to cart | Able to select product quantity before checkout |
| User interface design | Simple but not user-friendly interface | Simple user interface but basic UI design | Simple user interface but not choice of colors | Simple and user-friendly interface |
| Cart feature | Able to add, drop, and edit quantity of products in the cart | Able to drop and edit quantity of products in the cart | Cart feature unavailable | Able to add, delete, checkout, and select quantity of products in the cart |
| Order Analytics | Not available | Not available | Not available | Farmers can track order analytics, including daily, monthly, and weekly completed and pending orders |

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*D. Training the model*

During the prototyping stage, an interactive application prototype was created using the requirements gathered and shown to test users. This approach provided clear visibility into product browsing, order management, user roles, and navigation flow.

Based on the input obtained during the demonstration, changes were made to enable early resolution of usability concerns. These issues were addressed prior to moving on to the development stage thanks to the iterative prototype method.

*E. Testing the model*

During testing, system dependability and performance are ensured. The refined prototype is transformed into a functional model and subjected to extensive testing. Testing methods such as user acceptance testing (UAT), unit testing, and integration testing are used to identify and address defects. The system is continuously enhanced and adjusted in response to test results in order to achieve optimal performance.

*F. Implementation*

Farmers and local market vendors can access the finished application throughout the installation phase. The purpose of user acceptance testing (UAT) is to confirm that the system works as intended and to collect user input for any necessary final adjustments. Before the system is deployed, this iterative process guarantees that it satisfies all user needs. The development environment includes SQL for database administration, Spring Boot for backend services, and React Native for creating mobile applications. Order processing, inventory control, payment integration, and authentication are all handled by the backend APIs. Users can use their mobile number or email address to authenticate.

## III. RESULT

The planned digital platform's implementation has greatly improved agricultural trade by removing middlemen, guaranteeing fair pricing, and directly connecting farmers with local market merchants. By using Orders analytics and real-time price discovery, farmers can make well-informed decisions that optimize their sales tactics and boost profitability. The platform's use of automated logistics, inventory management, and secure payment methods has improved market accessibility, decreased post-harvest losses, and streamlined supply chain operations. In summary, the technology has increased market efficiency, given farmers access to real-time market information, and helped create a more lucrative and sustainable agricultural ecosystem.

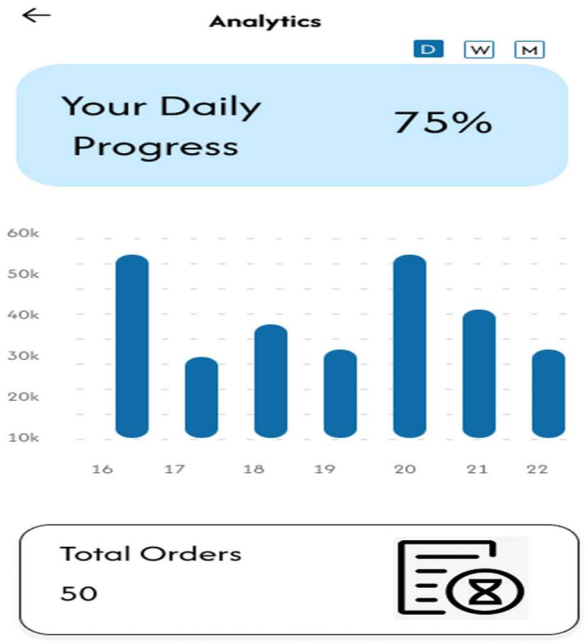



Fig. 2 Order Analytics

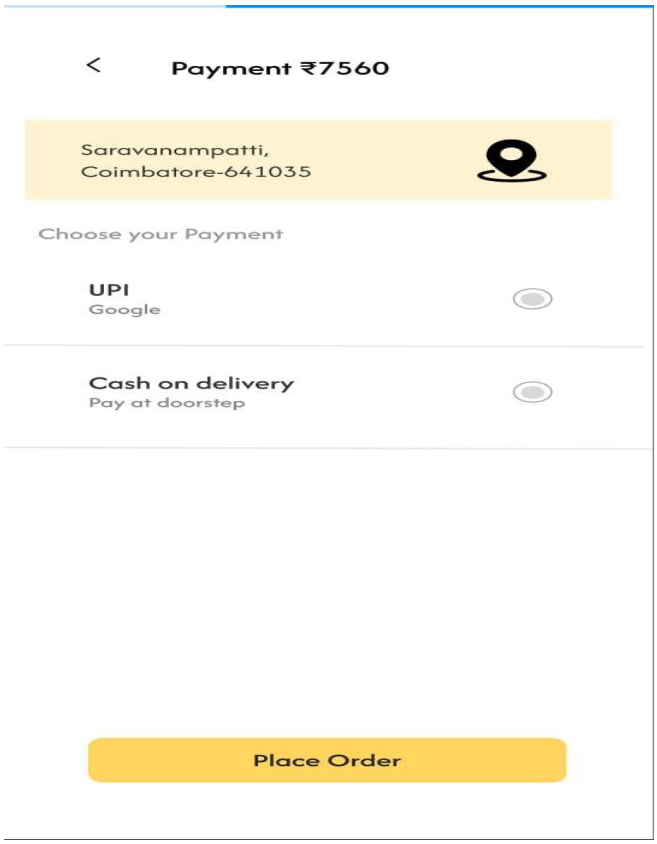



Fig. 3 Payment Option

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## CONCLUSION

In order to help streamline the agricultural supply chain, this project aims to create a digital platform that connects farmers and local market vendors. The system eliminates middlemen, provides real-time pricing, and uses predictive analytics to help farmers make better decisions, boost profitability, and improve market accessibility. The platform increases supply chain transparency, lowers post-harvest losses, and expedites trade procedures by leveraging contemporary technologies like React Native, Spring Boot, and SQL. By providing a more fair, open, and effective method of market access, the suggested solution has great potential to revolutionize agricultural trade. This digital platform highlights the significance of digital transformation and continuous innovation in contemporary agriculture by laying the groundwork for a more lucrative and sustainable agricultural ecosystem.

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